RIG-CLOSED POINTS IN THE IMAGE OF A FLAT ADIC FINITE TYPE MAP

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3418-3441, following the statement at 3400-3416.

Editorial note. The '(some details omitted)' at the reduction step is the standard preservation bookkeeping (adic, topologically finite type, flat under the quotient); flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3422 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Rig-closed primes and the relative conclusions. lemma-rig-closed-point-relative (same chapter, cited at ~3415 and ~3439-3440). Supplies the conclusions (varphi, p, q) referenced at the end.
- Nonzerodivisor nonunits on positive dimensional fibres. Algebra, Lemma algebra-lemma-dim-not-zero-exists-nonzerodivisor-nonunit (cross-chapter, cited at ~3433-3434). Produces b in B mapping to a nonzerodivisor nonunit of B_0.
- Flatness of the quotient by a nonzerodivisor. Algebra, Lemma algebra-lemma-grothendieck (cross-chapter, cited at ~3435-3436). B' = B/bB stays flat over A.

Source reference. Original reference 0GEC: focused-seeds/geometry/S9-restricted-rig-closed-point-in-image/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00MF: focused-seeds/geometry/S9-restricted-rig-closed-point-in-image/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GGB: sources/stacks-restricted/pinned-original-source.tex, source lines 3333–3333. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\varphi : A \to B$ be an arrow of $WAdm^{Noeth}$ which is adic and topologically of finite type. Let $\mathfrak{p} \subset A$ be rig-closed. Let $\mathfrak{a} \subset A$ and $\mathfrak{b} \subset B$ be the ideals of topologically nilpotent elements. If φ is flat, then the following are equivalent*

(1) *the maximal ideal of $A/\mathfrak{p}$ is in the image of* $\mathrm{Spec}(B/\mathfrak{b}) \to \mathrm{Spec}(A/\mathfrak{a})$,
(2) *there exists a rig-closed prime ideal $\mathfrak{q} \subset B$ such that $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$.*

and if so then φ, $\mathfrak{p}$, and $\mathfrak{q}$ satisfy the conclusions of Lemma 0GGB.

Proof. The implication (2) $\Rightarrow$ (1) is immediate. Assume (1). To prove the existence of $\mathfrak{q}$ we may replace A by $A/\mathfrak{p}$ and B by $B/\mathfrak{p}B$ (some details omitted). Thus we may assume $(A, \mathfrak{m}, \kappa)$ is a local complete 1-dimensional Noetherian ring, $\mathfrak{m} = \mathfrak{a}$, and $\mathfrak{p} = (0)$. Condition (1) just says that $B_0 = B \otimes_A \kappa = B/\mathfrak{m}B = B/\mathfrak{a}B$ is nonzero. Note that B_0 is of finite type over κ. Hence we can use induction on $\dim(B_0)$. If $\dim(B_0) = 0$, then any minimal prime $\mathfrak{q} \subset B$ will do (flatness of $A \to B$ insures that $\mathfrak{q}$ will lie over $\mathfrak{p} = (0)$). If $\dim(B_0) > 0$ then we can find an element $b \in B$ which maps to an element $b_0 \in B_0$ which is a nonzerodivisor and a nonunit, see Algebra, Lemma 0GEC. By Algebra, Lemma 00MF the ring $B' = B/bB$ is flat over A. Since $B'_0 = B' \otimes_A \kappa = B_0/(b_0)$ is not zero, we may apply the induction hypothesis to B' and conclude. The final statement of the lemma is clear from Lemma 0GGB. □